

SMART INDIA HACKATHON 2026 — OFFICIAL NOMINATION

AURA SENTINEL

AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

- **Team Name & ID:** ByteForce_1 (Official SIH Team ID: 180219)
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- **Problem Statement ID:** SIH26105 | Ministry of Education Innovation Cell (MIC) & AICTE Cyber Security Cell
- **Theme & Category:** Blockchain & Cybersecurity (Software Solution)
- **Core Mission:** Correlating technical security telemetry with business asset value to provide continuous financial exposure estimates and budget-constrained defense optimization.

Slide 02 | The Problem: The Disconnect in Cyber Risk Communication

CURRENT INDUSTRY LIMITATIONS

- ✗ **Qualitative Risk Ratings:** Vulnerability scanners produce labels like 'High', 'Medium', 'Low' which fail to convey actual monetary liability to leadership.
- ✗ **Stale Risk Registers:** Traditional compliance relies on annual or quarterly manual audits, missing dynamic threat evolution and emerging vulnerabilities.
- ✗ **Suboptimal Budget Allocation:** Organizations invest in security controls without quantifiable Return on Security Investment (ROSI) metrics.
- ✗ **Log Vulnerability:** Centralized audit logs in traditional databases remain susceptible to internal modification during a breach investigation.

THE AURA SENTINEL APPROACH

- ✓ **Monetary Cyber Risk Translation:** Translates real-time socket exposures, process states, and compute telemetry into prototype Expected Annual Loss (EAL) in ₹ INR.
- ✓ **Continuous Telemetry Correlation:** Continuously ingests local host metrics and open attack surfaces rather than relying on static questionnaires.
- ✓ **Budget-Constrained Optimization:** Evaluates candidate defense controls under an explicit institutional budget to maximize risk reduction.
- ✓ **Tamper-Evident Audit Ledger:** Anchors risk evaluations and mitigation actions in a SHA-256 Merkle blockchain to provide cryptographic audit verification.

Slide 03 | Core System Workflow: Technical Signals to Executive Decision

1. Technical Telemetry

Host metrics, listening sockets, CPU/RAM stress, and sensitive file exposures.

2. Continuous Analysis

Correlates exposure counts with system vulnerability factors and threat indicators.

3. Prototype Exposure

Computes hourly operational downtime costs and prototype Expected Annual Loss (₹).

4. What-If Scenarios

Simulates threat scenarios and hypothetical control parameter tuning.

5. Knapsack Optimizer

Selects optimal defense packages under explicit budget constraints to maximize ROSI.

6. Cryptographic Audit

Records assessments and mitigations into a tamper-evident SHA-256 Merkle ledger.

Slide 04 | Technical Architecture & Implementation Classification

LAYER 1: PRESENTATION & SOC

- React 19 + Vite SOC Dashboard
Status: ● IMPLEMENTED
- Live Threat Telemetry & Port Monitor
Status: ● IMPLEMENTED
- Global Command Palette (Ctrl+K)
Status: ● IMPLEMENTED
- Executive Audit Report Generator
Status: ● IMPLEMENTED
- Client Biometric Face Verification
Status: ● PROTOTYPE
- Procedural Cyber Audio Cues
Status: ● IMPLEMENTED

LAYER 2: COMPUTATIONAL CORE

- FastAPI Async Kernel (Uvicorn)
Status: ● IMPLEMENTED
- Attack Surface Port Inspector
Status: ● IMPLEMENTED
- Sensitive Credential Scanner
Status: ● IMPLEMENTED
- File Isolation Quarantine Vault
Status: ● IMPLEMENTED
- Prototype Financial Exposure (EAL)
Status: ● PROTOTYPE
- Knapsack Investment Optimizer
Status: ● PROTOTYPE
- What-If Scenario Simulation Engine
Status: ● IMPLEMENTED

LAYER 3: TRUST & INTERACTION

- SHA-256 Merkle Risk Ledger
Status: ● IMPLEMENTED
- Proof-of-Work Mining Engine
Status: ● IMPLEMENTED
- Tamper Detection & Verification
Status: ● IMPLEMENTED
- SQLite 3 WAL Mode Storage Hub
Status: ● IMPLEMENTED
- YOLOv8 Operator Presence Lock
Status: ● IMPLEMENTED
- MediaPipe Touchless Gesture Control
Status: ● IMPLEMENTED
- Value at Risk (Monte Carlo VaR)
Status: ○ FUTURE

Slide 05 | Attack Surface Telemetry & Risk Scoring

HOST TELEMETRY INGESTION (IMPLEMENTED)

- **Active Socket Inspection:** Queries network socket table via psutil, isolating listening ports and tracking bound IP interfaces.
- **Port Threat Categorization:** Heuristically maps sensitive ports (e.g. 22 SSH, 3389 RDP, 445 SMB) vs standard application ports.
- **Resource Stress Telemetry:** Monitors CPU percent, virtual memory usage, and disk saturation as stability and availability factors.
- **Active Port Remediation:** Includes prototype capability to terminate rogue listening processes or suppress non-critical noise.

WEIGHTED RISK SCORING MODEL

Technical Risk Score = $f(\text{Open Ports, CPU, RAM, Disk})$

- ✓ **Port Exposure Factor:** Up to 30 points based on listening socket count (>15 ports = +30, >8 ports = +20, >3 ports = +10).
- ✓ **Compute Stress Factor:** Up to 25 points for sustained CPU utilization (>90% = +25, >75% = +15).
- ✓ **Memory Exhaustion Factor:** Up to 25 points for virtual memory saturation (>90% = +25, >80% = +15).
- ✓ **Storage Saturation Factor:** Up to 20 points for critical disk capacity (>95% = +20, >85% = +12).
- ✓ **Threshold Normalization:** Clamped to 0–100 scale, categorizing system state into LOW, MODERATE, or HIGH risk.

Slide 06 | Prototype Financial Risk & Exposure Modeling (FAIR Grounded)

PROTOTYPE FORMULATION IN ₹ INR

- **Base Operational Loss Rate:** Calibrated at ₹15,000/hr as an illustrative enterprise baseline rate.
- **Risk-Adjusted Hourly Loss:** Hourly Loss = ₹15,000 × (1.0 + Technical_Risk_Score / 100).
- **Estimated Downtime Duration:** Dynamic mapping based on severity: HIGH = 8 hrs, MODERATE = 4 hrs, LOW = 1 hr.
- **Incident Direct Cost:** Direct Cost = Risk-Adjusted Hourly Loss × Downtime Hours.
- **Recovery Overhead:** Recovery Cost = Direct Cost × 0.35 (forensics & incident cleanup).
- **Total Estimated Exposure:** Total = Direct Incident Cost + Recovery Overhead + Asset Impact Factor.

FAIR ALIGNMENT & FUTURE MODELING

✓ Factor Analysis of Information Risk (FAIR):

The Open FAIR framework standardizes decomposing risk into Threat Event Frequency (TEF) and Loss Magnitude (LM).

✓ Prototype Implementation Status:

🕒 PROTOTYPE — Uses deterministic actuarial equations based on live host signals and user-supplied asset valuations.

✓ Value at Risk (VaR) Distinction:

○ FUTURE ENHANCEMENT — Monte Carlo probabilistic VaR distribution (e.g. 95% annual confidence) is scoped for future cloud telemetry integration.

✓ Defensible Evaluator Positioning:

Figures represent illustrative calculated risk exposure under current host conditions, not guaranteed financial forecasts.

Slide 07 | Decision Support: What-If Scenario Simulation Engine

HYPOTHETICAL TUNING (IMPLEMENTED)

- **Interactive Parameter Sliders:** Allows security operators to model varying CPU, memory, disk, and exposed port counts.
- **Live Exposure Re-calculation:** Instantly computes simulated risk score and projected monetary exposure as parameters shift.
- **Remediation Impact Assessment:** Demonstrates how reducing exposed ports from 18 to 2 reduces simulated business exposure.
- **Single-Click Live Sync:** Features button to pull actual live host metrics into the simulator as a baseline reference.

MITRE ATT&CK WARGAME SCENARIOS

- ✂ **SSH / RDP Brute Force Storm (T1110):** Simulates credential-stuffing pressure against remote daemons, generating live security alert logs.
- ✂ **Distributed SYN Flood DDoS (T1499):** Models volumetric connection exhaustion and elevated risk scoring.
- ✂ **Aggressive Reconnaissance Scan (T1046):** Simulates full-range port sweeps attempting to fingerprint local services.
- ✂ **Interactive Defense Trigger:** Operator triggers simulated mitigation, logging the defensive event into the persistent SQLite database.

Slide 08 | Budget-Constrained Security Investment Optimization

KNAPSACK OPTIMIZATION LOGIC

- 1. Input Parameters:**
Monthly Budget (e.g. ₹50,000 – ₹1,00,000), Managed System Count, and Critical Data Value (₹).
- 2. Candidate Defense Packages:**
Evaluates defined control tiers:
 - Essential (₹25,000/mo, ~15% Risk Reduction)
 - Professional (₹75,000/mo, ~35% Risk Reduction)
 - Enterprise (₹1,50,000/mo, ~60% Risk Reduction).
- 3. Budget Feasibility Filter:**
Constraint: $\text{Package_Cost} \leq \text{Stated_Monthly_Budget}$.
- 4. Return on Security Investment (ROSI):**
 $\text{ROSI (\%)} = [(\text{Potential Savings} - \text{Package Cost}) / \text{Package Cost}] \times 100$.

DECISION SUPPORT OUTPUT

- ✓ **Ranked Recommendation Score:**
Calculates composite score based on affordability (+40), positive net benefit (+30), and scaled ROI (+30).
- ✓ **Automated Plan Selection:**
Recommends the affordable tier delivering highest net financial savings within budget.
- ✓ **Tailored Security Actions:**
Generates prioritized recommendations (e.g. MFA enforcement, offline backup verification, port closure).
- ✓ **Prototype Status Disclosure:**
 - 🕒 **PROTOTYPE** — Illustrates algorithmic budget allocation; real enterprise deployment would ingest specific vendor RFP quotes.

Slide 09 | Trust Layer: Tamper-Evident SHA-256 Merkle Audit Ledger

CRYPTOGRAPHIC LEDGER MECHANISM

- **Security Event Trigger:** Risk assessments, threat quarantines, and budget changes emit structured payloads.
- **Merkle Root Calculation:** Pairwise SHA-256 hashing condenses arbitrary payload keys into a single cryptographic Merkle root.
- **Proof-of-Work Mining:** Calculates candidate header hash with dynamic nonce until satisfying target difficulty prefix.
- **Chained Block Header:** $\text{Block Hash} = \text{SHA256}(\text{Index} + \text{Timestamp} + \text{Event} + \text{MerkleRoot} + \text{PrevHash} + \text{Nonce})$.
- **Defensible Terminology:** Described as a 'Tamper-Evident Audit Ledger' — unauthorized modifications break cryptographic chain verification.

VERIFICATION & TAMPER DEMO

- ✓ **Full Chain Validation (verify_integrity):**
Traverses chain from Genesis (Block 0) to tip, verifying previous_hash links, recomputed Merkle roots, and block hashes.
- ✓ **Simulate Tamper Attack (DEMO):**
Modifies stored data payload of Block #1 in JSON storage to simulate rogue log alteration.
- ✓ **Instant Cryptographic Detection:**
Next integrity audit immediately detects Merkle root mismatch and broken hash chain, flagging the exact invalid block.
- ✓ **Consensus State Restoration:**
Restores verified snapshot from backup memory pool, returning ledger to verified state.

Slide 10 | Supporting Endpoint Capabilities: Physical Presence & Interaction

ZERO-TRUST PHYSICAL AUTO-LOCK

- **Problem Addressed:** Physical workstation access remains a frequent insider threat vector if an operator walks away leaving a console unlocked.
- **YOLOv8 Operator Tracking:** Ultralytics YOLOv8-nano monitors operator presence in local webcam feed.
- **Configurable Grace Period:** If operator departs, a countdown banner alerts the user before executing OS-level LockWorkStation call.
- **Supporting Role in SIH:** Presented accurately as an endpoint hardening feature, secondary to the core financial risk quantification engine.

TOUCHLESS NEURAL GESTURE CONTROL

✓ MediaPipe Hand Landmarks:

Tracks 21 3D hand coordinates locally using Google MediaPipe Hand Landmarker.

✓ Deterministic Vector Logic:

Crossed fingers (Index over Middle) detected via coordinate inversion vector ($mcp_diff \times tip_diff < 0$).

✓ Gesture Capabilities:

- 📌 Index Point: New Folder hotkey
- ✌️ Crossed Fingers: File delete action
- 🖐️ Pinch & Slide: System volume control
- 🕊️ Peace Sign: Application window switch.

✓ Architecture Positioning:

Designed to run locally on CPU hardware without requiring external cloud GPU servers.

Slide 11 | Regulatory Framework Alignment & Control Mapping

INDIAN REGULATORY CONTEXT

-  **CERT-In Cyber Security Directions (April 2022):**
Mandates 180-day preservation of cybersecurity incident logs. AURA's blockchain ledger provides cryptographic evidence of non-alteration.
-  **RBI Cyber Security Framework:**
Supports risk threshold monitoring, sensitive port alerting, and cyber resilience disclosures for institutional banking environments.
-  **SEBI Cybersecurity Resilience Framework:**
Addresses board-level risk metrics, technical asset inventory tracking, and systematic risk governance.
-  **DPDP Act 2023 Compliance Support:**
Regex credential scanner flags unmasked Aadhaar, PAN numbers, and private keys in scanned directories.

GLOBAL FRAMEWORK MAPPING

-  **NIST Cybersecurity Framework 2.0:**
Structured mapping across core functions:
 - Identify: Asset & port inventory
 - Protect: Defense knapsack optimizer
 - Detect: Socket & compute telemetry
 - Respond / Recover: Quarantine vault & audit.
-  **ISO/IEC 27001 (A.12 Operations Security):**
Aligns with technical control logging and vulnerability management clauses.
-  **CIS Controls v8:**
Supports Control 1 (Enterprise Assets), Control 4 (Secure Configuration), and Control 8 (Audit Log Management).
-  **Defensible Claim Note:**
Framed accurately as 'control alignment and compliance-supporting capabilities', not official government certifications.

Slide 12 | Proposed Architecture: Future Multi-Institution Deployment

PROPOSED ARCHITECTURE FOR FUTURE MULTI-INSTITUTION DEPLOYMENT (CONCEPTUAL SCALING)

NODE 01: CENTRAL GOVERNANCE NODE

Proposed Central Authority (e.g. Consortium Master)

- Aggregates anonymized risk exposure indexes across institutions
- Sets baseline risk evaluation standards
- Distributes global threat intelligence advisories
- Acts as primary consensus coordinator

NODE 02: REGIONAL VALIDATOR NODE

Proposed Regional Validator (e.g. Lead Technical Campus)

- Validates consensus blocks via independent Merkle checks
- Executes regional peer risk verification
- Distributes computational load for consortium mining
- Maintains verifiable mirror of consortium ledger

NODE 03: INSTITUTIONAL AUDITOR NODE

Proposed Campus Auditor (e.g. Individual College Node)

- Operates local-first agent monitoring campus endpoints
- Emits tamper-evident risk assessments to the ledger
- Validates compliance posture against regulatory requirements
- Prevents local insider repudiation of breach incidents

Slide 13 | Implementation Status Matrix & Development Roadmap

RUNNABLE IN CURRENT MVP (TODAY)

- ✓ **Host Telemetry & Port Inspector:** Live extraction of open sockets, CPU/RAM utilization, and process states (psutil).
- ✓ **Prototype Financial Exposure (EAL):** Live computation of hourly downtime loss and estimated incident cost in ₹ INR.
- ✓ **What-If Simulation Engine:** Interactive parameter tuning and MITRE attack vector wargaming.
- ✓ **Budget Investment Optimizer:** Knapsack evaluation of defense packages, ROI %, and tailored action list.
- ✓ **Blockchain Ledger & Tamper Demo:** SHA-256 blocks with Merkle roots, Proof-of-Work mining, and live tamper simulation.
- ✓ **Physical Security & Gestures:** YOLOv8 operator auto-lock + MediaPipe touchless OS mouse controls.

FUTURE PRODUCTION ROADMAP

- 🚀 **Enterprise Ingestion Connectors (Q1):**
Native log forwarders for Wazuh, Splunk, CrowdStrike EDR, and AWS/Azure CSPM.
- 🚀 **Monte Carlo Probabilistic VaR (Q2):**
Statistical probability density distributions estimating Value at Risk at 95% confidence intervals.
- 🚀 **Multi-Campus Consortium Network (Q2):**
Deploying real peer-to-peer validator nodes across participating universities.
- 🚀 **Automated Dynamic Control Discovery (Q3):**
Direct API integration with IT procurement systems for dynamic security solution pricing.
- 🚀 **Full DPDP Audit Automation (Q3):**
Automated compliance reports formatted for direct submission to Indian regulatory authorities.

Slide 14 | Evaluator-Grounded 5-Minute Live Demonstration Sequence

STEP-BY-STEP LIVE DEMO FLOW (100% DEMONSTRABLE ON CURRENT BUILD)

- 🕒 **0:00 – 0:45 | Dashboard & Host Status:** Launch AURA SOC Dashboard (Module 01). Show live compute telemetry, listening socket counts, and normalized technical risk score.
- 🕒 **0:45 – 1:30 | Attack Surface Detection:** Navigate to Attack Surface (Module 02). Highlight active listening ports, danger classifications, and process IDs.
- 🕒 **1:30 – 2:15 | Prototype Financial Risk (EAL):** Open Financial Risk (Module 04). Demonstrate live translation of socket stress into hourly downtime rate and estimated exposure in ₹ INR.
- 🕒 **2:15 – 3:15 | What-If Scenario Simulation:** Open What-If Engine (Module 05). Adjust sliders to demonstrate exposure reduction; trigger MITRE wargame scenario and log mitigation.
- 🕒 **3:15 – 4:15 | Budget Investment Optimizer:** Open Investment Optimizer (Module 06). Input ₹1,00,000 budget; execute Knapsack optimizer to display recommended tier and ROSI %.
- 🕒 **4:15 – 4:45 | Blockchain Tamper Showdown:** Open Blockchain Ledger (Module 10). Run integrity audit (100% valid) → Trigger tamper simulation → Show instant red alert → Restore consensus.
- 🕒 **4:45 – 5:00 | Executive Audit Report Wrap-up:** Click 'Audit Report' button in navbar. Present printable executive report summarizing compliance alignment and actuarial exposure.